

Mathematics Standard

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your student number and/or name at the top of every page

This paper MUST NOT be removed from the examination room

STUDENT NUMBER/NAME:

Total marks – 100

Section I – Pages 2–11

20 marks

- Attempt Questions 1–20
- Allow about 30 minutes for this section

Section II – Pages 12–28

80 marks

- Attempt Questions 21–43
- Allow about 1 hour and 30 minutes for this section

STUDENT NUMBER/NAME:

Section I

20 marks

Attempt Questions 1–20

Allow about 30 minutes for this section

Select the alternative A, B, C or D that best answers the question and indicate your choice with a cross (X) in the appropriate space on the grid below.

	A	B	C	D
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20				

1 What is the mode of the data set: {8, 9, 12, 15, 15}?

- A. 7
- B. 11.8
- C. 12
- D. 15

2 The time-and-a-half rate paid to employees of a company is \$72 per hour.

What would be the double-time rate per hour paid to the employees?

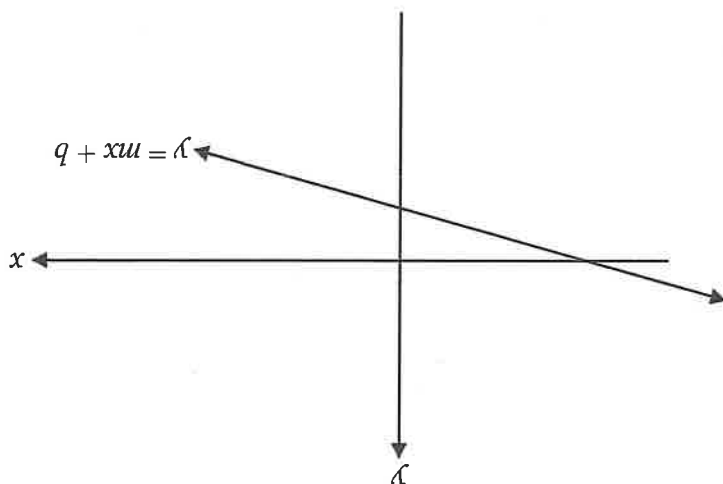
- A. \$48
- B. \$96
- C. \$108
- D. \$144

3 City B is 20° West and 30° South of City A.

If City A has co-ordinates (25°N, 5°E), what are the co-ordinates of City B?

- A. (0°, 45°E)
- B. (5°S, 15°W)
- C. (15°S, 5°E)
- D. (5°N, 15°W)

4 A line with equation $y = mx + b$ is shown on the grid below.



Which statement is correct of the values of m and b in the equation of the line?

- A. m is negative and b is positive.
- B. m is positive and b is negative.
- C. Both m and b are negative.
- D. Both m and b are positive.

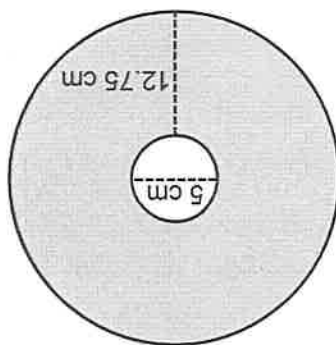
5 The probability that a ten pin bowling competitor will score over 200 in any game is 0.85.

What is the probability that the competitor will score over 200 in at least one of the next two games?

- A. $1 - (0.15)^2$
- B. $(0.85)^2$
- C. $0.85 + 0.85$
- D. $1 - (0.85)^2$

6 An annulus consists of two concentric circles, as shown in the diagram below.

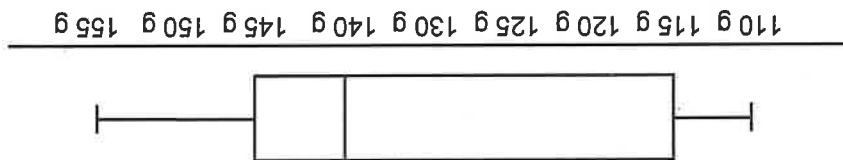
DIAGRAM NOT
TO SCALE



What is the approximate area (cm^2) contained between the two concentric circles of the annulus shown?

- A. 432
- B. 491
- C. 652
- D. 711

7 500 packets of grain were weighed and it was found that the contents varied between 110 g to 155 g, as shown in the box-and-whisker plot below.



How many packets of grain had weights recorded between 140g and 145g?

- A. 5
- B. Less than 60
- C. 125
- D. More than 250

8

19 out of 20 students sat an assessment task producing a mean of 78%.

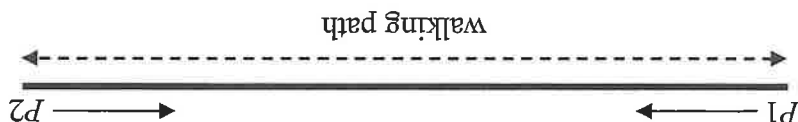
When the 20th student sat the task the next day, the mean increased to 78.5%.

What result did the 20th student achieve on the task?

- A. 79%
- B. 80%
- C. 82.5%
- D. 88%

9

Two people, P_1 and P_2 , begin walking from opposite ends of a straight path.



P_1 leaves one end of the path at 10:15 am and walks at an average speed of 6 km/h, arriving at the other end of the path at 10:55 am.

P_2 leaves the opposite end of the path at 10:30 am and arrives at the other end of the path at 11:00 am.

What was the average speed (km/h) of person P_2 ?

- A. 4.5
- B. 5.5
- C. 6
- D. 8

10

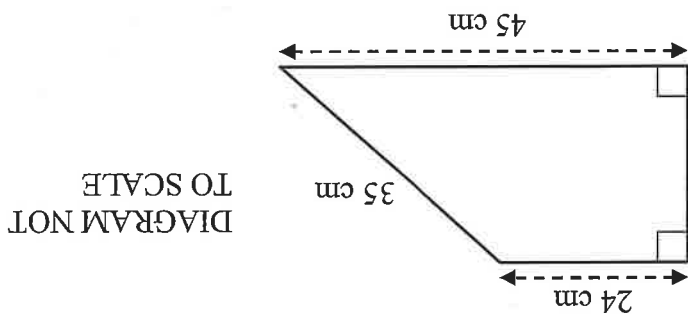
The formula: $T = G - \frac{50k}{7}$ can be used to determine the temperature T , outside a plane flying at k kilometres above the ground, where G is the ground temperature.

The temperature outside the plane is -38° when the ground temperature is 22° .

At what height (km) is the plane flying above the ground?

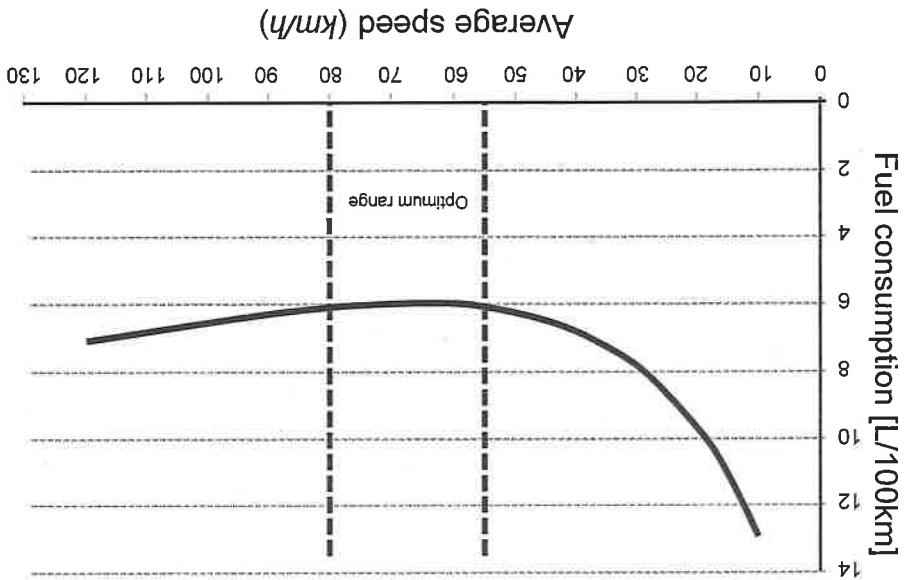
- A. 7.25
- B. 8
- C. 8.4
- D. 8.25

11 What is the area (cm^2) of the trapezium shown below?



- A. 654
- B. 787.5
- C. 966
- D. 1207.5

12 Janet is considering purchasing a new car and examines a graph of the car's published fuel consumption figures for various speeds.

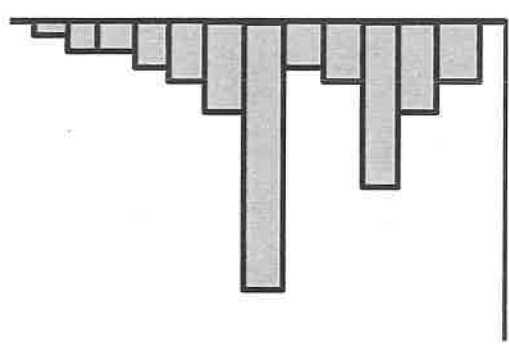


The car has a 60-litre fuel tank.

On a full tank of fuel, approximately what distance (km) should the car be expected to travel according to the data presented in the graph, if travelling within the optimal average speed range?

- A. 850
- B. 900
- C. 950
- D. 1000

13 The following diagram shows a data distribution



Which of the following correctly describes the distribution?

- A. Unimodal and negatively skewed.
- B. Unimodal and positively skewed.
- C. Bimodal and negatively skewed.
- D. Bimodal and positively skewed.

14 Which expression gives P as the subject of the formula $\bar{Q} = \frac{2P^2}{5}$?

- A. $\frac{\sqrt{5Q}}{2}$
- B. $\frac{5\sqrt{Q}}{2}$
- C. $\sqrt{\frac{5Q}{2}}$
- D. $5\sqrt{\frac{Q}{2}}$

- 15 A tape is used to measure the distance a student jumps in a long jump event at a school athletics carnival.
- What would be the percentage error in a recorded measurement of 3.65 metres in this event?

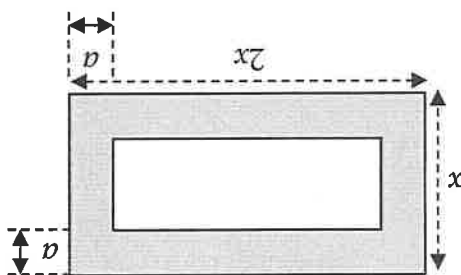
A. 0.14
 B. 0.27
 C. 0.5
 D. 1.4

- 16 A petrol tank of a truck has a capacity of 90 litres.

What would be the approximate volume (cubic metres) of the truck's petrol tank?

A. 9×10^{-4}
 B. 9×10^{-3}
 C. 9×10^{-2}
 D. 9×10^{-1}

- 17 A rectangular piece of cardboard of length $2x$ cm and width x cm has a border of width a cm shaded, as shown in the diagram.



What is the perimeter (cm) of the inner rectangle?

A. $2(x - 4a)$
 B. $2(3x - 4a)$
 C. $2(2x - a)$
 D. $2(x + 4a)$

- 18 The following table indicates the number of residents from four age groups in a community who have the latest version of a mobile phone.

Age group (years)	Number of residents
18–25	68 500
26–40	55 200
41–60	78 000
Over 60	33 300

For research, an online stratified survey of 20 000 people from these age groups was conducted to determine whether there were any issues with the new phones. Approximately how many people would be asked to complete the survey from the 41–60 age group?

- A. 780
B. 1560
C. 6638
D. 15 600

- 19 Dubai has a Universal Coordinated Time (UTC) of +4 hours and Sydney has a UTC of +10 hours.

A plane leaves Sydney at 6:00 am Monday morning and travels to Dubai with a flight time of approximately 13 hours.

At what local time in Dubai would the plane be expected to arrive?

- A. 10:00 am Monday
B. Midday Monday
C. 1:00 pm Monday
D. 3:00 pm Monday

20 Based on alcohol consumption and body weight, the Blood Alcohol Content (BAC) of an average male and female can be calculated by use of the formulae:

$$BAC_{\text{Male}} = \frac{6.8M}{10N - 7.5H} \quad BAC_{\text{Female}} = \frac{5.5M}{10N - 7.5H}$$

where N is the number of standard drinks consumed, H is the number of hours drinking and M is the person's weight in kilograms.

Two patrons of a club (one male and one female) have both consumed 3 standard alcoholic drinks within 2 hours and weigh approximately the same.

If the male recorded a BAC of 0.045, what would be the female's BAC?

- A. 0.056
- B. 0.059
- C. 0.060
- D. 0.063

Section II

80 marks

Attempt Questions 21–43

Allow about 1 hour and 30 minutes for this section

Answer the questions in the spaces provided, using a black pen.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the end of the examination. If you use this space, clearly indicate which question you are answering.

Marks

Question 21 (1 mark)

If $y = 3^x + 3^{-x}$ what is the value of y if $x = 2$?

1

Question 22 (2 marks)

Calculate the volume of a sphere of diameter 5.5 cm.

2

(Use the formula $V = \frac{4}{3}\pi r^3$, correct to one decimal place.)

Question 23 (2 marks)

Solve the equation: $-4m - 3 = 6 - m$.

2

Question 24 (1 mark)**Marks**

The exchange rate between the Australian dollar and the United States dollar is:

1

$$\text{\$AUD } 1.00 = \text{\$USD } 0.764$$

An Australian tourist in the United States makes a purchase for \$USD 850.

What is the equivalent amount in Australian Dollars?

Question 25 (2 marks)

Calculate the value of $\sqrt{\frac{4\pi^3}{5}}$, correct to two decimal places.

2

Question 26 (3 marks)

A table of scores with cumulative frequencies is shown below.

Score	Cumulative frequency
15	7
25	10
35	18
45	34
55	40

(a) Which score represents the mode for the presented data? 1

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(b) What is the mean? 2

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Question 27 (3 marks)

The circumference of Mars is approximately 2.13×10^4 km.

3

The surface area (S) of a sphere is given by the formula: $S = 4\pi r^2$, where r is the radius.
 What is the approximate surface area of Mars (square kilometres)?
 (Give your answer in scientific notation correct to 3 significant figures)

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Question 28 (6 marks)

The number of sit ups completed in a set period by members of a gym are shown in the stem-and-leaf plot below.

2	5 5 8 9
3	0 0 0 2 4 5 8 8 9
4	0 0 2 2 3 5

(a) What is the range in the number of sit ups completed? 1

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(b) What is the median number of sit ups completed? 1

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(c) Write a five-number summary for the data presented. 2

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(d) Describe the distribution of the data presented. 2

Justify your answer with mathematical reasoning.

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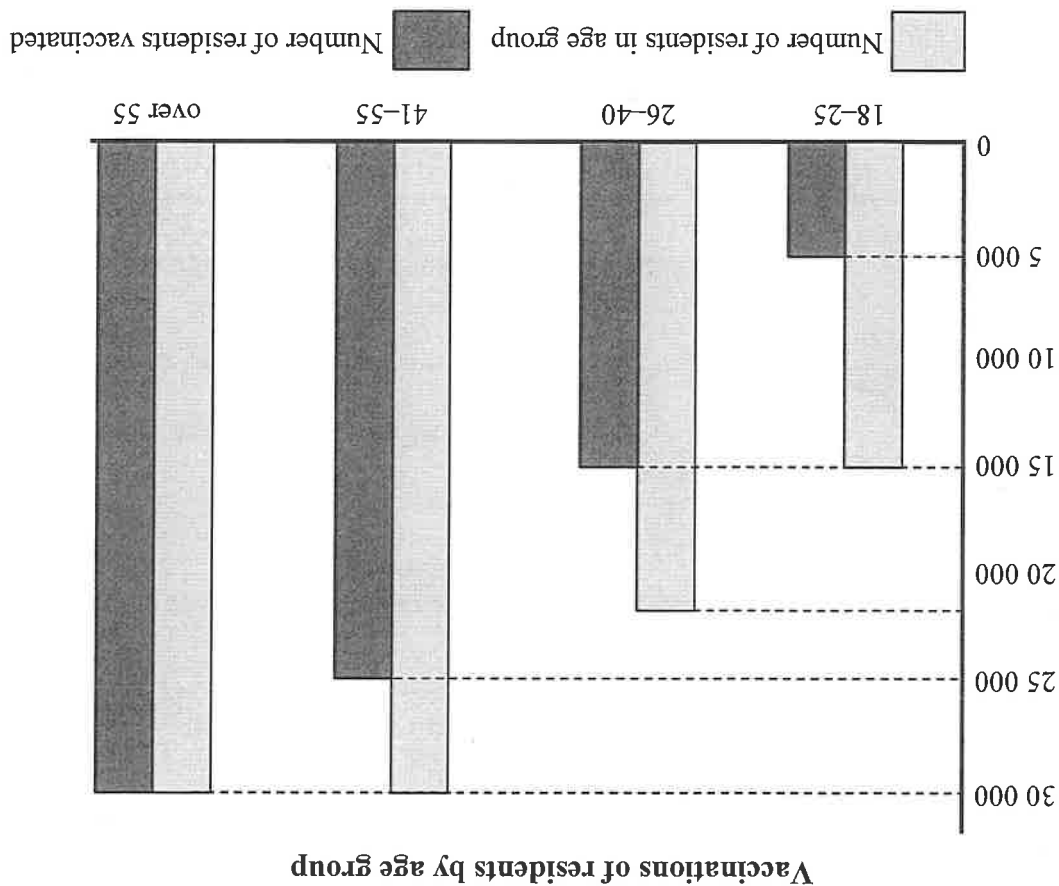
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STUDENT NUMBER/NAME:

Question 29 (5 marks)**Marks**

The graph below displays data collected from a community on the number of residents in different age groups who have been vaccinated for protection against a virus.



(a) Which age group has the smallest percentage of residents who have been vaccinated? **1**

(b) What percentage of the residents in the 41-55 age group have been vaccinated? **1**

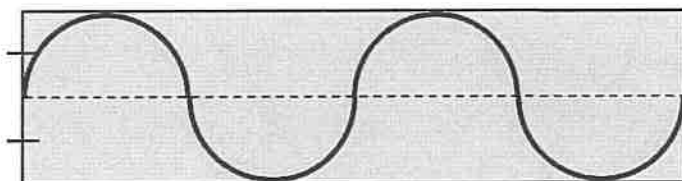
(c) What trend is evident from the data in the graph? **1**

(d) If two residents were randomly chosen from the 18-25 age group, what is the probability that at least one of these residents has been vaccinated? **2**

Marks

3

A piece of string of length 48 cm is formed into 4 semicircles of equal radius and placed on top of a rectangular piece of cardboard so that it touches the edges of the cardboard, as shown.



What is the perimeter of the cardboard, to the nearest centimetre?

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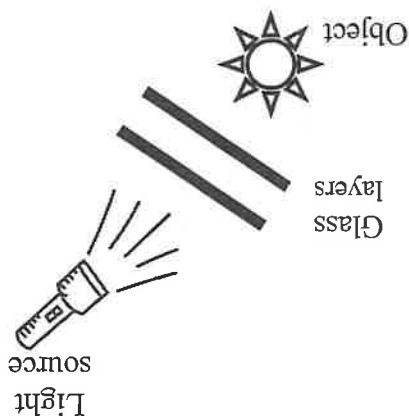
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Question 31 (2 marks)

2

Each layer of glass reduces the amount of light from a source onto an object by 12%.



Determine the percentage loss of light from the source through two layers of glass, as shown in the diagram.

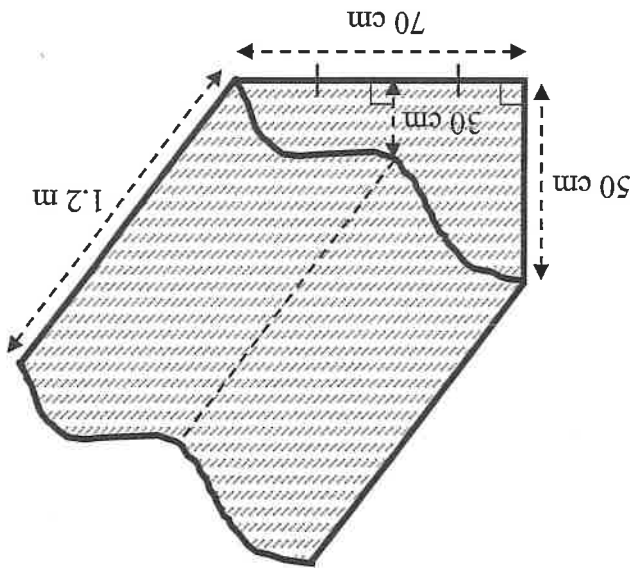
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Marks

3 A piece of foam rubber has been cut into the shape below for a customer order.



Using two applications of the trapezoidal rule, calculate the volume of the piece of foam in cubic metres, correct to one decimal place.

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Question 33 (3 marks)

3 An annual flat rate of 3.25% interest is paid on investments of \$10 000 or more at a bank.

After 5 years, an investment with this bank has a value of \$14 531.25.

What amount over \$10 000 was invested 5 years ago?

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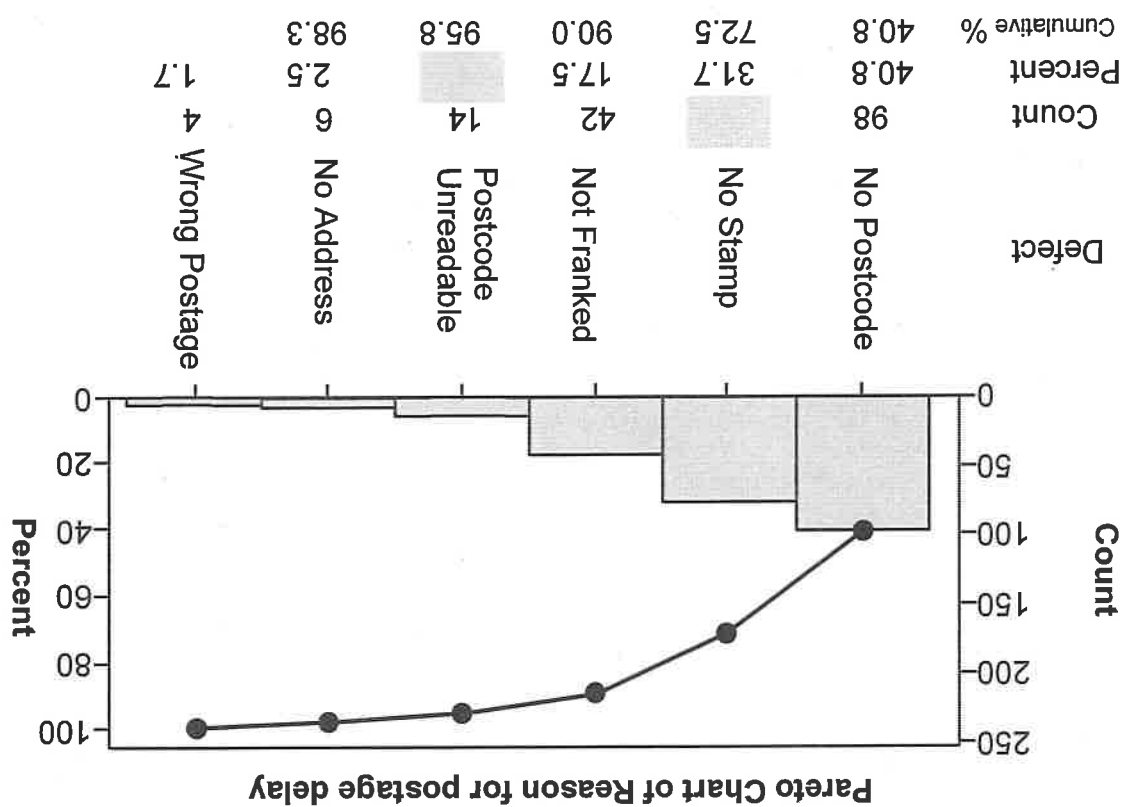
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Marks

2

A Pareto chart indicates the reasons given by a post office for postal delays.

Question 34 (2 marks)



The figures in the table the chart give the data relating to each reason for the postal delays.

Complete the two missing figures in the shaded boxes of the table above, correct to one decimal place.

Some animals are twice as likely to have male offspring (M) as female offspring (F).

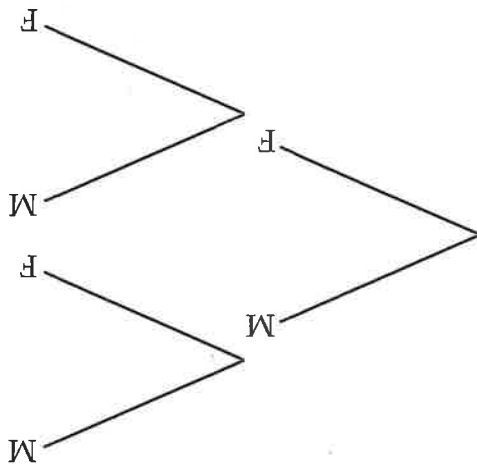
- (a) Explain why the probability of one of these animals having a male is $\frac{2}{3}$. 1

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- (b) One animal is expected to have two offspring.

The following tree diagram shows the outcomes.



- (i) Complete the tree diagram inserting the probabilities of each outcome. 2

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- (ii) What is the probability that this animal will have a female and a male offspring? 1

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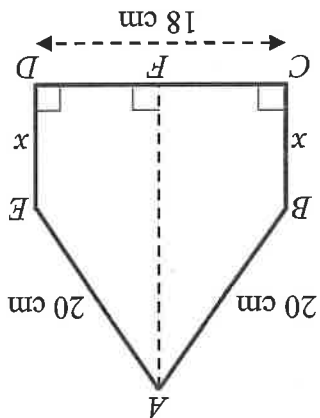
- (iii) Justify the statement "It is more likely that this animal will have both babies of the same sex, than a male and a female offspring." 1

Marks

Question 36 (5 marks)

A piece of wire is bent to form a pentagon $ABCDE$.

F is the point on CD so that AF is perpendicular to CD , as shown.



(a) If the wire has length 88 cm, what is the value of x ?

2

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(b) Calculate the area of the pentagon.

3

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Question 37 (3 marks)

Jayden receives a total gross amount of \$10 394.24 paid in advance of his upcoming 4 weeks leave from his job, which includes a holiday loading of 17.5% of 4 weeks gross salary.

3

What annual salary does Jayden earn?

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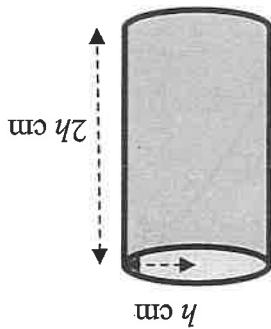
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Question 38 (7 marks)

Marks

The following diagram shows a closed cylinder (with base and top surfaces) with height $2h$ cm and radius h cm.



(a) Show that the surface area of the cylinder is $6\pi h^2$ cm².

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(b) If the surface area of the cylinder is 96π cm², show that the volume of the cylinder is approximately 402 cm³.

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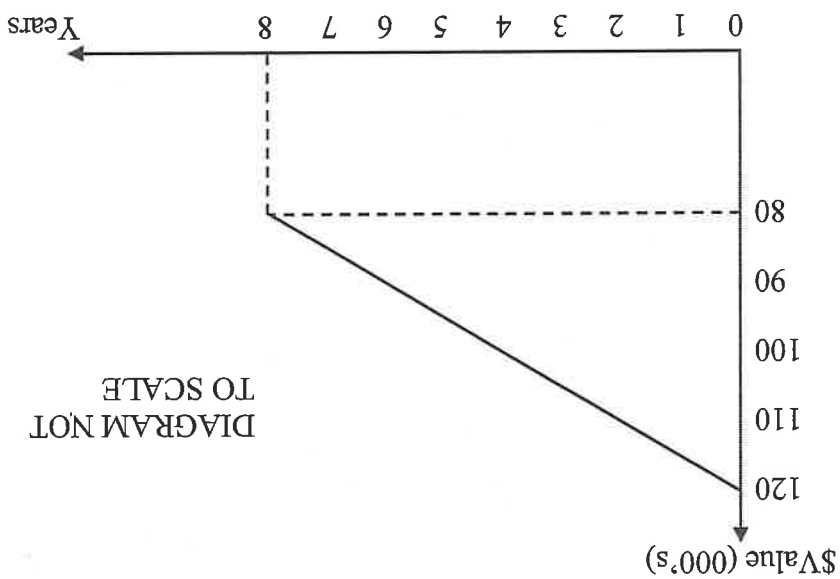
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Marks

Question 39 (5 marks)

STUDENT NUMBER/NAME:

The graph shows the depreciated value of a factory machine over a period of 8 years.



(a) What annual rate of depreciation is indicated by the information on the graph?

(b) What will be the machine's total loss in value after a further 2 years?

(c) When is the machine estimated to reach a value of zero?

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Question 40 (6 marks)

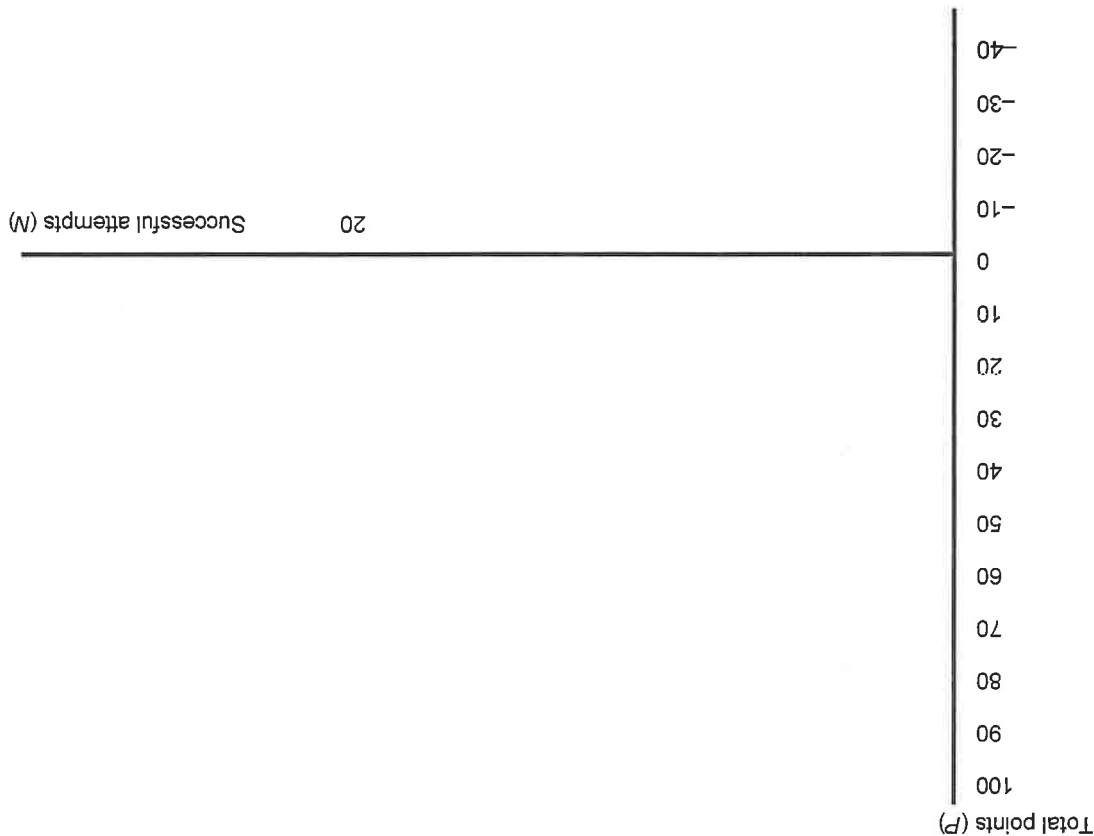
Marks

In a competition, players have 20 shots at scoring goals in each round. 3 points are awarded for each successful attempt at scoring a goal and 1 point is deducted for each unsuccessful attempt at scoring a goal. The player or players with the least number of points, do not progress to the next round of the competition.

(a) What is the maximum and minimum number of points a player can score in this competition? 2

(i) Maximum:
 (ii) Minimum:

(b) On the grid below, draw a linear graph which would represent the number of points P that can be scored with N successful attempts at a goal. 2



(c) By writing and solving the equation representing the graph, determine the number of successful attempts at a goal for which a player would score a total of zero points. 2

Question 41 (6 marks)

STUDENT NUMBER/NAME:

- (a) One method of identifying an outlier in a data set, is to consider a score which lies outside the range:

Mean - (3 × sample standard deviation) to (Mean + 3 × sample standard deviation)

Use this method to show that the score of 25 would not be considered an outlier in the data set:

25, 45, 46, 46, 52, 55, 58, 60, 64

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- (b) A second method of identifying an outlier in a data set is to consider a score which lies outside the range:

$$Q_1 - (1.5 \times IQR) \text{ to } (Q_3 + 1.5 \times IQR)$$

Use this method to show that the score of 25 in the above data set could be considered an outlier.

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- (c) Give an explanation as to why these two methods give different outcomes and which method would be more reliable.

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Question 42 (2 marks)

When a tradesperson adds 10% Good and Services Tax (GST) to a client's invoice for

work completed, the client pays \$*D* to the tradesperson.

What amount in terms of *D*, was the amount on the invoice before the GST was added?

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Question 43 (3 marks)

Jason used the tax table below to calculate that the tax payable on his taxable income for the financial year was \$21 342.

3

0 – \$18 200	Nil
\$18 201 – \$45 000	19 cents for each \$1 over \$18 200
\$45 001 – \$120 000	\$5092 plus 32.5 cents for each \$1 over \$45 000
\$120 201 – \$180 000	\$29 467 plus 37 cents for each \$1 over \$120 000
\$180 001 and over	\$51 667 plus 45 cents for each \$1 over 180 000

What was Jason's taxable income?

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End of paper

Section II – Extra writing space

STUDENT NUMBER/NAME:

Section II – Extra writing space

STUDENT NUMBER/NAME: